JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Electrical Engineering
"Igniting minds to illuminate the world"

VISION	MISSION
"To develop competent and committed Electrical Engineers to serve the society"	1. To impart quality education in the field of Electrical Engineering. 2. To be excellent learning center through research and industry interaction.

Program: B.Tech in Electrical Engineering

Semester	Course Code	Name of the course	L	T	P	Credits
		Solar Technologies	3	0	0	3

Prerequisites for the course	
1	UG level understanding of MQC,IED.

Prior Reading Material/useful links	
1	https://www.youtube.com/watch?v=c3CalfdYZYw&list=PLpe3qgeJLpB2wAoaRSY9_yAeOt7u0LTNd
2	https://www.youtube.com/watch?v=qfkTVYJIx8Q&list=PLpe3qgeJLpB2wAoaRSY9_yAeOt7u0LTNd&index=2

Course Outcomes:

Sr.No	Course Outcome number	CO statement
1	CO1	**Define** the basics of solar energy.
2	CO2	**Describe** the various applications utilizing the solar thermal system.
3	CO3	**Compute** the dimensions for the solar energy system.
4	CO4	**Apply** fundamentals of solar energy to modern-day applications.

Syllabus:

Course Contents	
Unit I	**Basics of solar energy** **[08Hrs]** Energy Scenario, what is solar energy, Generation of solar energy ,physics of propagation of solar radiation from the sun to earth, heat to electricity conversion.
Unit II	**Solar thermal systems** **[08Hrs]** Fundamentals of solar collectors, Liquid flat plate collectors and concentrating collectors. Solar cookers, Solar Air heaters. Solar thermal power generation. Solar pond. Solar refrigeration, Passive architecture, solar distillation, and emerging technologies.
Unit III	**Solar Photovoltaic system** **[08Hrs]** Fundamentals of solar PV cells, principles and performance analysis, modules, arrays, theoretical maximum power generation from PV cells. Solar water pumping, Satellite. Solar Street lights, Solar mobile chargers. Solar rechargeable lamps. PV based farming applications.
Unit IV	**Solar-electric vehicles** **[08Hrs]** Solar cars. Solar buses. Single-track vehicles. Auxiliary power. Personal rapid transit. Rail. They manned solar aircraft.
Text Books	
1	Sukhatme,S.P.,&Nayak,J.K.(2017).Solarenergy.McGraw-HillEducation.
2	Rai,G.D.(2011).A Textbook of Non-conventional energy sources.
3	Garg, H.P.(2000).Solar energy: fundamentals and applications. Tata McGraw-Hill Education.
	Tyagi,H.,Agarwal,A.K.,Chakraborty,P.R.,&Powar,S.(2018).Introduction To applications of solar energy. In Applications of Solar Energy (pp.3-10). Springer, Singapore.
Reference Books	
1	Reinders, A., Verlinden, P.,vanSark, W.(2017).Photovoltaic Solar Energy: From Fundamentals to Applications. United Kingdom: Wiley.
	Boxwell,M.(2010).Solar Electricity Handbook: ASimple, Practical Guide to Solar Energy – Designing and Installing Photovoltaic Solar Electric Systems. United Kingdom: Green stream Publishing.
Useful links	
1	https://www.youtube.com/watch?v=c3CalfdYZYw&list=PLpe3qgeJLpB2wAoaRSY9_yAeOt7u0LTNd
2	https://www.youtube.com/watch?v=qfkTVYJIx8Q&list=PLpe3qgeJLpB2wAoaRSY9_yAeOt7u0LTNd&index=2

Contributions for syllabus designing:

Sr. No	Name of the person	Designation	Organization
1	Dr.V.S.Dhok	Asst.Professor	JDCOEM,Nagpur
2	Mr.A.V.Joshi	Asst.Professor	JDCOEM,Nagpur
3	Ms.P.P.Panchbhai	Asst.Professor	JDCOEM,Nagpur
4	Mr.P.V.Ambade	Asst.Professor	JDCOEM,Nagpur
5	Mr.M.S.Isasare	Asst.Professor	JDCOEM,Nagpur
6	Mr.Rishabh Darwai	Systems Engineer	Infosys Systems ,Pvt.Ltd,Bengaluru

Mr. A.V.Joshi,
Member Secretary,

Dr. V.S.Dhok
Chairperson